MILITARY SPECIFICATION

TEST SET, ALTIMETER WARNING AN/APM-254

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The test set covered by this specification shall be for shop and flightline use, and shall be used to test, calibrate, and troubleshoot the Radar Altimeter Warning Set AN/APQ-107.

1.2 Classification - The test set covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Test Set, Altimeter Warning TS-2508/APM-254	3.5.1
Case, Test Set, CY-6095/APM-254	3.5.2
Cable Assembly, Power, W-1	3.5.3
Cable Assembly, Signal, W-2	3.5.3
Cable Assembly, Auxiliary Power, W-3	3.5.3
Hose Assembly, Altitude	3.5.4
Chart, Checkout	3.5.5

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-R-7722	Radar Set, AN/APN-22
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for

FCS-6625

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MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-21079	Radar Set, AN/APN-117
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

DRAWINGS

Naval Air Systems Command

DL 66A10D1	Data List for Test Set, Altimeter Warning AN/APM-254, and all drawings listed therein

PUBLICATIONS

Military

ANA Bulletin No. 400	Electronic Equipment: Aircraft and Guided Missiles, Applicable Documents

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications and documents required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENT

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials -

3.2.1 General Requirements - The parts and materials specified in the drawings and related documents for use in construction of this equipment were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change, and the cost associated therewith, upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipments for preproduction tests. Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of the performance requirements or any other requirements of this specification.

3.2.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained by the contractor as outlined in MIL-T-21200.

3.2.2 Supplemental Requirements - In the event that the contractor proposes to alter the equipment design delineated by this specification and applicable drawings, the following supplemental

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requirements for the selection of parts and materials are applicable.

3.2.2.1 Basic Conformance - Parts and materials selected for use in the equipment shall conform to the requirements of MIL-T-21200. Parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform adequately in the intended applications and environments specified herein.

3.2.2.2 Interchangeability - The test set shall meet the interchangeability requirements of MIL-T-21200. The drawings listed in DL 66A10D1 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.3 Design and Construction - The test set shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 66A10D1 and related lists, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case it is a condition of final acceptance that the equipment manufactured meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically approved by the procuring activity prior to their incorporation.

3.3.1 Total Weight - The total weight of the test set, including cables, shall not exceed 35 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The test set shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for adjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The test set shall have a minimum total operating life of 20,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.3.3 Radio Interference - The generation of radio interference by the test set and the vulnerability of the test set to radio interference shall be within the limits of MIL-I-6181, except that short duration interference resulting from manual operation of switches shall be within the relaxed limits of MIL-I-6181 as follows:

(1) For interference with a maximum duration of 1 second and a maximum recurrence of once in 3 minutes, a 20 db deviation from the limits is permitted.

(2) For interference with a maximum duration of 3 seconds and a maximum recurrence of twice for normal operational period, there is no limitation on the deviation from the limits.

3.3.4 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 volts, 400 cps a-c, ± 5 cps 25 ± 0.5 volts d-c

3.3.5 Service Conditions - The test set shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for class 2 equipment, except as modified herein.

3.3.5.1 Vibration - The test set shall operate satisfactorily when subjected to the vibration test of MIL-T-21200 for the frequency range of 5 to 25 cps. For the frequency range of 25 to 55 cps, the following exceptions to full performance are permissible:

(1) The stability (or accuracy) of the barometric altimeter/altitude chamber shall be within ± 50 feet of the accuracy specified for standard conditions.

(2) The barometric altitude chamber control knob shall not be required to maintain

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its position except by manual control.

Following vibration the barometric altitude chamber shall operate satisfactorily, and within standard tolerances.

3.3.5.2 Normal Mounting Position - The normal mounting position of the test set for service and standard conditions shall be with the front panel horizontal and the case resting on the four bottom feet.

3.3.6 Warm-Up Time - The time required for the test set to warm-up prior to operations shall be kept to a minimum and shall not exceed 2 minutes under standard conditions and 10 minutes under extreme service conditions.

3.3.7 Primary Input Power Requirements - The test set shall operate satisfactorily when supplied with input electrical power as defined in MIL-T-21200, except that the power shall be within the following limits:

(1) A-C Power, Single Phase, 115 volts ± 10%, 380 to 420 cps, 115 VA.

(2) D-C Power, 27 ± 2.0 volts, 3.0 amperes.

3.4 Performance - The performance of the test set shall be as set forth below plus any and all performance requirements stated on the drawings of DL 66A10D1. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified.

3.4.1 Operation - The test set provides the following functions.

(1) Provides a means of simulating radar altimeter inputs to the Radar Altimeter Warning Set under test.

(2) Provides input power to the Radar Altimeter Warning Set under test.

(3) Provides a means of simulating the various enabling and disabling inputs required to test the Radar Altimeter Warning Set.

(4) Monitors the warning and delayed reliability outputs from the Radar Altimeter Warning Set.

(5) Provides a means of simulating barometric altitudes from normal ground to 7000 feet above normal ground to test the barometric switch in the Radar Altimeter Warning Set.

(6) Provides self-test capabilities.

3.5 Detail Requirements -

3.5.1 Test Set, Altimeter Warning TS-2508/APM-254 - The test set shall be manufactured in accordance with the documents listed on DL 66A10D1, shall meet the requirements stated therein, and the following requirements:

3.5.1.1 Function - The test set shall perform the following functions:

3.5.1.1.1 Radar Altitude Simulation - The test set shall simulate the altitude as a function of output signal voltages from the following radar altimeters:

(1) Altimeter Set, Electronic AN/APN-141(V) - The test set shall simulate the output signal voltages (measured from pins J1-h and J1-j to J1-T (d-c ground) from this altimeter in accordance with the following equations:

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$V = E_1A + C_1$, $0 \leq A \leq 400$ feet

$V = E_2A + C_2$, $400 \leq A \leq 500$ feet

where: E_1 = 2.00 volts per 100 feet

E_2 = 2.61 volts per 1000 feet

C_1 = 1.00 volt

C_2 = 7.95 volts

The altitude signals shall be within an accuracy of ±5 percent.

(2) Radar Set AN/APN-117 (Radar Altimeter) - The test set shall simulate the synchro voltages (measured from pins J1-X and J1-Y to J1-T) from the altimeter by a self-contained 15TR4 type synchro torque transmitter and a control dial calibrated in accordance with figure 1 of MIL-R-21079.

(3) Radar Set AN/APN-22 (Radar Altimeter) - The test set shall simulate the synchro voltages (measured from pins J1-X and J1-Y to J1-T) from the altimeter by a self-contained 15TR4 type synchro torque transmitter and a control dial calibrated to an altitude of 1000 feet in accordance with figure 1 of MIL-R-7722. The requirement of this and the preceding paragraph shall be satisfied by a single synchro control transmitter and a dual-calibrated control dial.

3.5.1.1.2 Power - The test set shall provide electrical power to operate the Radar Altimeter Warning Set as follows:

A-C Power: Single Phase, 115 volts ± 10%, 380 to 420 cps, 20 VA (measured from pin J1-b to J1-c).

D-C Power: 27 ± 2.0 volts d-c at 0.2 ampere (measured from pin J1-J to J1-T).

3.5.1.1.3 Aircraft Signal Simulated Outputs -

(1) Pilot Radar Altimeter Limit Indicator Signal - The test set shall provide a 7 ± 2-volt, 400 ± 20-cps a-c, and a +27 ± 2-volt d-c signal (both measured from pin J1-n to J1-T) to simulate the ON condition, and a d-c ground signal to simulate the OFF condition of the pilot limit indicator. These signals shall conform to these values when applied to a 24,000-ohm resistance load.

(2) Copilot Radar Altimeter Limit Indicator Signal - The test set shall provide the signals described in 3.5.1.1.3(1) (measured from pin J1-p to J1-T) to simulate the ON and OFF conditions of the copilot limit indicator when the signal is applied to a 24,000-ohm resistance load.

(3) Altimeter Indicator Power Signal - The test set shall provide a 115-volt ± 15%, 400 ± 20-cps signal (measured from pin J1-M to J1-T) to simulate the ON condition, and zero volts to simulate the OFF condition of the radar altimeter.

(4) Reliability Signal - The test set shall provide a +4.3 ± 0.3-volt d-c signal (measured from pin J1-E to J1-T) when externally loaded with 10,000 ohms resistance to simulate the Positive Reliability mode, and zero volts to simulate the Negative Reliability mode of the radar altimeter.

(5) Test Command Signal - The test set shall provide a +27 ± 2 volt-d-c signal (measured from pin J1-R to J1-T) to energize the self test circuit of the AN/APQ-107. The signal shall be open circuited in the OFF position.

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(6) ASE Engagement Signal - The test set shall provide a +27 ± 2 volt-d-c signal (measured from pin J1-D to J1-T) to simulate the engaged condition, and a zero-volt signal to simulate the disengaged condition of the ASE coupler.

(7) Weight-On-Wheels Signal - The test set shall provide a d-c ground signal (measured from pin J1-q to J1-T) to simulate the ON condition, and an open circuit to simulate the OFF condition.

(8) Wheel Down and Locked Signals - The test set shall provide three independently-controlled signals (measured from pins J1-B, J1-C, and J1-Z to J1-T) of +4.3 ± 0.3 volts d-c when externally loaded with 7500 ohms resistance to simulate the UP position of the wheels, and d-c ground to simulate the DOWN position.

(9) Synchro Excitation - The test set shall provide a 115-volt ± 15%, 400 ± 20 cps signal (measured from pin J1-V to J1-T) to simulate the synchro excitation voltage from the AN/APN-22 and AN/APN-117 radar altimeters. An open circuit signal shall be provided to simulate the no-voltage condition.

(10) Pressure Altitude Test Signal - The test set shall provide a d-c ground signal (measured from pin J1-N to J1-T) to simulate closing, and an open circuit to simulate opening of the barometric switch located in the AN/APQ-107.

(11) High Altitude Warning Inhibit - The test set shall provide a +27 ± 2-volt d-c signal (measured from pin J1-k to J1-T) to simulate the high altitude inhibit signal engaged, and a ground signal to simulate the high altitude inhibit signal disengaged.

3.5.1.1.4 Monitor Requirements -

(1) Delayed Reliability Signal - The test set shall provide a visual indicator (from pin J1-P to J1-T) with a load simulating 100 ± 20 milliamperes to monitor the delayed reliability output from the AN/APQ-107.

(2) Visual Warning - The test set shall provide a visual indicator (from pin J1-F to J1-T) with a load simulating three (MS25237-327) lamps connected in parallel to monitor the visual warning signals from the AN/APQ-107.

(3) Pilot and Copilot Audio - The test set shall provide a means of selecting either the pilot or copilot audio signals (from pin J1-G to J1-H, and from J1-K to J1-L) from the AN/APQ-107 and applying them either to the speaker or to a 750 ± 40-ohm resistance load.

(4) +28 VDC Test - The test set shall provide a visual indicator (from pin J1-A to J1-T) in series with a single-pole push button switch to monitor the +28 volt d-c power from the AN/APQ-107.

(5) Test Points - The test set provides test points for monitoring the following signals from the AN/APQ-107:

(a) +28 volt d-c power supply

(b) -20 volt d-c power supply

(c) delayed reliability signal

(d) synchro converter (low altitude signal)

(e) synchro converter (high altitude signal)

(f) visual warning signal

(g) pilot and copilot audio warning signal

(h) wheel inhibit signal

(i) ground

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(6) Barometric Altitude - The barometric altimeter shall conform to 66A10D52. The barometric altimeter monitors the barometric pressure simulated by the active chamber and indicates from zero to 7500 feet minimum.

3.5.1.1.5 Barometric Altitude Simulation Requirement - The test set shall simulate barometric altitude from normal ground to 7000 feet.

3.5.1.2 Form Factor - The test set shall be contained in a combination case conforming to 66A10F21.

3.5.1.3 Weight - The weight of the test set shall not exceed 30 pounds.

3.5.2 Case, Test Set, CY-6095/APM-254 - The test set case shall be manufactured in accordance with the documents listed on 66A10F21.

3.5.3 Cable Assemblies - Cable assemblies shall be manufactured in accordance with drawings as follows:

Item	Part No. (dwg)
Cable Assembly, Power W-1	66A10D31
Cable Assembly, Signal W-2	66A10D41
Cable Assembly, Auxilliary Power W-3	66A10D51

3.5.3.1 Functions (Descriptive) - The cable assemblies provide functions as described below:

3.5.3.1.1 Cable Assembly, Power W-1 - Cable W-1 applies input power from the shop bench to the test set.

3.5.3.1.2 Cable Assembly, Signal W-2 - Cable W-2 applies all signals from the test set to the AN/APQ-107 unit under test.

3.5.3.1.3 Cable Assembly, Auxiliary Power W-3 - Cable W-3 applies input power from the aircraft to the test set when the test set is used on the flightline.

3.5.4 Hose Assembly, Altitude - The altitude hose assembly shall be manufactured in accordance with the documents listed on 66A10C91 and shall meet the requirements stated thereon.

3.5.4.1 Function (Descriptive) - The altitude hose assembly connects the simulated altitude to the barometric switch in the AN/APQ-107.

3.5.5 Chart, Checkout - The checkout chart shall be manufactured in accordance with the documents listed on 60A10F61 and shall meet the requirement stated thereon.

3.5.5.1 Function (Descriptive) - The checkout chart provides simplified test instructions for flightline testing.

3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see 6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental and interference tests in accordance with the requirements of MIL-T-5422 to the limits specified in MIL-T-21200.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be

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cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Vibration Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Vibration Test - Each equipment shall be vibrated (without vibration isolators) under the following conditions for a period of 10 minutes.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. A record shall be kept of all failures caused by vibration.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual, and special tests.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (see 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Tests procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual tests to check all equipments for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Tests, Sampling

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Tests, Reliability Tests, or Special Tests may be selected for Life Tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room ambient (30 ± 10°C)
Altitude	Normal ground (0 to 5000 ft.)
Humidity	Room ambient
Input Voltage	115 ± 10% (at lowest applicable frequency) 27.0 ± 2.0 volts d-c
Vibration	None

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor

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by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipments - Equipments which have been subjected to acceptance tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipments for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipments which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the originals shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - The Altimeter Warning Test Set is intended to perform operational tests, calibration, and fault isolation of the Radar Altimeter Warning Set AN/APQ-107 either in the shop or on the flightline.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement of materials, parts, and processes other than those specified in DL 66A10D1, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information, including the requirements of 3.3.2.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) The drawings of DL 66A10D1 - The drawings shall have precedence over other

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referenced specifications. Any deviation from the drawings shall be specifically approved by the procuring activity.

(4) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Parentheses - The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with MIL-N-18307. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

6.7 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81570(AS) TEST SET, ALTIMETER WARNING AN/APM-254

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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